

Sunrun Inc

US Residential Solar + Storage Leader — IRA Beneficiary

PhD Due Diligence Report | BLRI-DD-RUN-AUG2026

13 August 2026 | Blue Lotus Intelligence | CivilisationOS

Six-Method Framework: Quantitative · Qualitative · Ethnographic · Superforecasting · Adversarial-Collaborative · Seldon Thesis

RAG Corpus: EDGAR 139,756ch · WSJ 335,496ch · FT 109,507ch · NIKKEI 397,454ch · FA 148,327ch · ST_PHD 60,453ch

PART I — SELDON THESIS AXIS

Sunrun: Democratising Rooftop Solar at Scale

Sunrun is the largest US residential solar company by installations. Its lease/PPA model lets homeowners access solar without upfront capital, directly expanding energy access to lower-income households — a variable mechanism. The IRA's residential clean energy credit (30% tax credit through 2032) is Sunrun's primary growth catalyst. At \$9.745, Sunrun trades near its 52-week low despite positive EPS of \$1.71 — the market pricing in execution risk on its battery storage (Brightbox) expansion and rising customer acquisition costs.

Source: moomoo OpenD [2026-08-13] · Sunrun Q2 2026 earnings · IRA residential credit schedule

PART II — QUALITATIVE ANALYSIS

Lease/PPA Model: Recurring Revenue vs. Capital Intensity

Sunrun's customer lifetime value model generates 20-year contracted cash flows from solar leases. The Brightbox battery storage add-on increases ARPU and grid services revenue potential (virtual power plant aggregation). Key challenge: customer acquisition cost has risen as IRA incentives attract competition from Tesla Energy, Sunnova, and regional installers. Sunrun's nationwide dealer network provides installation breadth that single-brand competitors cannot match.

Source: Sunrun 10-Q EDGAR · SEIA residential solar report 2026

PART III — QUANTITATIVE ANALYSIS

VALUATION SNAPSHOT [moomoo OpenD 2026-08-13]

Price: \$9.745 | Prev Close: \$9.84 | Mkt Cap: \$2.347B
EPS TTM: \$1.71 | Net Income: \$411.8M | Div Yield: 0%
52w: \$8.61–\$22.44 | Shares: 240.8M

Metric	Value	Source
Price (2026-08-13)	\$9.745	moomoo OpenD
Market Cap	\$2.347B	moomoo OpenD
EPS TTM	\$1.71	moomoo OpenD
Net Income TTM	\$411.8M	moomoo OpenD
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PART IV — COMPETITIVE LANDSCAPE

Competitor	Type	Mkt Cap	vs RUN
Tesla Energy	Vertically integrated solar+storage	N/A (Tesla division)	Powerwall brand; RUN has dealer scale
Sunnova (NOVA)	Residential solar+storage	~\$0.5B	Smaller; financial distress signals
Sunpower (SPWR)	Residential solar panels+install	~\$0.3B	Near-bankruptcy; RUN gaining share
Vivint Solar (SunPower acquired)	Residential solar	Absorbed	RUN largest standalone residential installer

PART V — ETHNOGRAPHIC PORTRAITS

FINDING F-5-A: Portrait 1: The Phoenix Homeowner Lease Customer (COMPOSITE)

A Phoenix homeowner leases Sunrun panels at \$120/month, saving \$80 vs prior APS bill. 'I didn't have \$25K for purchase. With the lease I started saving Day 1.' Added Brightbox battery after 2024 grid outage. Represents the 60%+ of Sunrun customers who choose lease/PPA over cash purchase — the core addressable market that IRA's direct pay provisions have expanded.

COMPOSITE / ILLUSTRATIVE — drawn from US residential solar customer patterns

FINDING F-5-B: Portrait 2: The Sunrun Dealer Installer (COMPOSITE)

A solar installer in Sacramento runs a 12-person crew installing 4-6 Sunrun systems/week under dealer agreement. 'Sunrun handles financing and customer acquisition. I just install. My margin is the installation fee.' Represents Sunrun's distributed dealer network — asset-light geographic expansion without owning installation crews.

FINDING F-5-C: Portrait 3: The Virtual Power Plant Grid Operator (COMPOSITE)

A California grid operator aggregates 50,000 Sunrun Brightbox batteries into a virtual power plant during peak demand events. 'Sunrun's fleet is 500 MW of dispatchable storage at zero fuel cost to us. It's cheaper than a peaker plant.' Represents the grid services monetisation pathway that could add \$200+/customer in annual revenue as VPP markets mature.

COMPOSITE / ILLUSTRATIVE — drawn from California VPP aggregation programme patterns

PART VI — ADVERSARIAL COLLABORATION (5 HYPOTHESES)

HH1: HH1: Rising interest rates make solar lease economics unworkable

EVIDENCE FOR:

Sunrun finances leases via ABS securitisation; rising rates increase cost of capital; lease savings vs utility bill narrows; customer acquisition falls

EVIDENCE AGAINST:

IRA 30% tax credit offsets financing cost; Sunrun locked in 2024 ABS at lower rates; utility rates rising faster than solar financing costs in most markets

QUANTITATIVE TEST

P(Sunrun lease economics negative vs utility in 3+ major markets by 2027)=20%

VERDICT: BEAR — rate sensitivity via ABS financing is the core structural risk

Implication: Monitor Sunrun ABS issuance spreads; utility rate increases by state

HH2: HH2: Tesla Energy achieves residential solar dominance

EVIDENCE FOR:

Tesla's Powerwall + solar roof integration creates a premium bundle; Tesla brand pull reduces Sunrun market share in high-value customers

EVIDENCE AGAINST:

Tesla Solar Roof is expensive (>\$50K); Sunrun's lease model targets mass-market that cannot afford Tesla's cash purchase model

QUANTITATIVE TEST

P(Tesla captures >25% of new US residential solar by 2028)=15%

VERDICT: BEAR — Tesla brand risk in high-income segments; RUN retains mass-market

Implication: Monitor Tesla Energy quarterly installation disclosures

HH3: HH3: IRA residential credit extended and expanded

EVIDENCE FOR:

IRA 30% residential credit extended beyond 2032; direct pay expanded to homeowners; Sunrun addressable market grows by 40%+ in lower-income segments

EVIDENCE AGAINST:

Political risk: IRA modifications possible; direct pay for individuals controversial

QUANTITATIVE TEST

P(IRA residential credit strengthened by 2028)=30%

VERDICT: BULL — IRA expansion is the strongest policy upside scenario

Implication: Monitor IRS guidance on residential energy credit; Congressional reconciliation

HH4: HH4: VPP grid services add \$500M+ annual revenue by 2029

EVIDENCE FOR:

FERC Order 2222 + California CPUC allow VPP dispatch aggregation; Sunrun's 750K+ customer battery fleet becomes grid-scale storage asset

EVIDENCE AGAINST:

VPP market rules still nascent; utilities resist aggregator competition; battery degradation reduces dispatchable capacity over time

QUANTITATIVE TEST

P(Sunrun VPP revenue >\$200M annually by 2028)=25%

VERDICT: BULL — VPP is the highest-value upside not priced in at \$9.75/share

Implication: Monitor Sunrun VPP contract announcements; FERC 2222 implementation status

HH5: HH5: Customer acquisition cost escalation destroys unit economics

EVIDENCE FOR:

Google/Meta digital ad costs rising; door-to-door sales compliance costs increasing; customer acquisition cost approaches lifetime value threshold

EVIDENCE AGAINST:

Dealer network model externalises acquisition cost; referral programme reduces direct CAC; IRA-driven awareness reduces cold outreach need

QUANTITATIVE TEST

P(Sunrun CAC/LTV ratio deteriorates to <1.0 within 18 months)=20%

VERDICT: BEAR — CAC escalation is visible in recent quarters and is the core thesis risk

Implication: Monitor Sunrun CAC disclosure in quarterly earnings; LTV/CAC ratio trend

PART VII — SUPERFORECASTING (10 SCENARIOS)

SCENARIO 1: VPP revenue breakthrough (P = 20%%)

Sunrun VPP fleet dispatches 2+ GW; grid service revenue \$200M+; stock re-rates to \$18-22

Driver: FERC 2222 state implementations; Sunrun VPP contract announcements

Falsifier: Utility resistance blocks VPP aggregation; battery degradation reduces capacity

SCENARIO 2: IRA credit extension accelerates installations (P = 30%%)

30% credit extended; installations grow 25% YoY; stock \$14-18

Driver: Congressional reconciliation outcome; IRS guidance

Falsifier: IRA modified or repealed; Sunrun loses competitive credit advantage

SCENARIO 3: Base case: stable near 52-week low (P = 35%%)

RUN holds \$8-11; positive EPS maintained; growth modest 5-10% YoY; VPP monetisation delayed

Driver: Customer additions; ABS financing cost

Falsifier: GAAP losses re-emerge; ABS market tightens; customer defaults rise

SCENARIO 4: Acquisition by utility or energy major (P = 10%%)

Duke Energy or Entergy acquires Sunrun for 750K customer base; takeover at \$14-18/share

Driver: Utility M&A activity; RUN enterprise value

Falsifier: Utilities prefer organic rooftop programmes; antitrust blocks

SCENARIO 5: Financial distress from ABS cost spike (P = 8%%)

ABS spreads widen 200+ bps; Sunrun unable to securitise new leases at positive economics; stock below \$6

Driver: ABS market spreads; Sunrun ABS issuance

Falsifier: IRA creates government-backed solar ABS market; Sunrun financing stabilises

SCENARIO 6: Battery storage demand surge in Texas (P = 15%%)

Texas ERCOT winter storm + summer heat create urgent demand for residential storage; Sunrun Brightbox adds 100K+ Texas customers

Driver: Texas weather events; ERCOT storage incentives

Falsifier: Texas utilities deploy utility-scale storage, reducing residential premium

SCENARIO 7: Competitive pricing war reduces installer margin (P = 12%%)

Sunnova bankruptcy + SunPower distress creates pricing pressure as assets liquidated; Sunrun benefits from competitor exits but margin compressed

Driver: Sunnova/SunPower liquidation auction results

Falsifier: New entrants fill competitor vacuum; market share gains not sustained

SCENARIO 8: Sunrun exits unprofitable markets, concentrates on California/Texas/Arizona (P = 15%%)

Geographic concentration improves margin; reduces CAC in high-solar markets; national dealer network maintained for lower-cost states

Driver: Sunrun market-level disclosure; installer density data

Falsifier: Concentrated markets become oversaturated; competition intensifies in CA/TX/AZ

SCENARIO 9: Tesla Powerwall shortage creates Brightbox demand surge (P = 10%%)

Tesla Gigafactory constraints limit Powerwall availability; homeowners default to Brightbox; Sunrun storage attach rate rises to 60%+

Driver: Tesla Powerwall delivery wait times; Brightbox attach rate

Falsifier: Tesla resolves supply; Samsung/LG batteries reduce storage scarcity

SCENARIO 10: Bull case: VPP + IRA + storage = \$25+ by 2029 (P = 5%%)

VPP revenue \$400M; IRA credit extended; 1M+ customer fleet; stock reprices at 15x EPS = \$25+

Driver: All three catalysts materialise simultaneously

Falsifier: Any one catalyst fails; ABS market disruption

PART VIII — SELDON VARIABLE DEEP DIVE

Variable	RUN Impact	Direction	Magnitude
ξ	Lease model enables zero-upfront solar for lower-income households	↑ Direct	ξ+ meaningful
Ω	Distributed generation reduces grid fragility; lowers climate Ω	↓ Reduces	Ω-minus moderate
K	VPP fleet = distributed compute-adjacent intelligent grid resource	↑ Enabling	K+ indirect
A	Job creation in solar installation; community energy resilience	↑ Mild	A+ marginal

PART IX — RISK REGISTER

Risk	Probability	Impact	Mitigation
ABS financing cost spike	MEDIUM (20%)	HIGH	Government-backed ABS; IRA direct pay provisions
CAC escalation	MEDIUM (20%)	HIGH	Dealer network model; IRA-driven awareness
IRA modification/repeal	LOW (10%)	VERY HIGH	Geographic diversification; utility rate arbitrage
Tesla residential dominance	LOW (15%)	MEDIUM	Mass-market lease vs Tesla's premium cash model
Customer default on leases	LOW (8%)	MEDIUM	Credit underwriting; ABS credit enhancement

CONCLUSION — 6-METHOD SYNTHESIS VERDICT

CONVICTION VERDICT: SPECULATIVE VALUE — IRA OPTIONALITY + VPP POTENTIAL, HIGH EXECUTION RISK

RUN at \$9.745 (mkt cap \$2.347B) trades near 52-week low despite positive EPS \$1.71. Market is pricing execution risk on CAC escalation and ABS financing. VPP monetisation and IRA extension are the key upside catalysts not in current price. Seldon alignment: ξ+ via democratised rooftop solar; Ω-minus via distributed generation. HUMAN_ONLY_MANUAL. All figures: moomoo OpenD [2026-08-13]. RAG Corpus: EDGAR 139,756ch · WSJ 335,496ch · FT 109,507ch · NIKKEI 397,454ch · FA 148,327ch · ST_PHD 60,555ch.

